SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 5

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Due on Nov. 3, 2026, 18:00 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

If $X \sim \text{HGeom}(w, b, n)$, what is the distribution of $n - X$? Give a short proof.

$X \sim HGeom(w,b,n)$. $P(X=k) = \dfrac{\binom{w}{k}\cdot\binom{b}{n-k}}{\binom{w+b}{n}}$

let $Y = n-X$, $X = n-Y$.

$P(Y=k) = P(X=n-k) = \dfrac{\binom{w}{n-k}\cdot\binom{b}{k}}{\binom{w+b}{n}}$, so that $n-X \sim HGeom(b, w, n)$

$\max(0, n-w) \le Y \le \min(n, b)$

Problem 2

A particular Youtuber is known for his arbitrary steak-eating habit during live streaming. In each live streaming day, he orders a steak with doneness as one of well done, medium well, medium, medium rare, and rare (with equal probability, independent of other live streaming days). How many days of live streaming do you expect to watch before you see him eating steaks with each possible doneness at least one once (suppose you are a big fan who watches every live streaming)?

Assume W_i : have seen $i-1$ doneness, waiting for ith doneness

the expect $T = W_1 + W_2 + W_3 + W_4 + W_5$.

the tth time for W_i to be successful :

$P(W=t) = (1-p)^{t-1} p$, $t=1,2,3,\cdots$, $E[W] = \frac{1}{p}$,

for W_i, $p = \frac{5-(i-1)}{5} = \frac{6-i}{5}$, $E[W_i] = \frac{5}{6-i}$.

$T = 1 + \frac{5}{4} + \frac{5}{3} + \frac{5}{2} + 5 = 6 + \frac{15+10+30}{12} = 6 + \frac{65}{12} = \frac{137}{12} \approx 11.42$ days

Problem 3

Mario and Zelda are independently performing independent Bernoulli trials. For concreteness, assume that Mario is flipping a nickel with probability p_1 of Heads and Zelda is flipping a penny with probability p_2 of Heads. Let $X_1, X_2, \cdots$ be Mario's results and $Y_1, Y_2, \cdots$ be Zelda's results, with $X_i \sim \text{Bern}(p_1)$ and $Y_j \sim \text{Bern}(p_2)$.

(a) Find the distribution and expected value of the first time at which they are simultaneously successful, *i.e.*, the smallest n such that $X_n = Y_n = 1$.

(b) Find the expected time until at least one has a success (including the success).

(c) For $p_1 = p_2$, find the probability that their first successes are simultaneous, and use this to find the probability that Mario's first success precedes Zelda's.

(a) $X_i \sim \text{Bern}(p_1)$, $Y_j \sim \text{Bern}(p_2)$

Assume T : they are simultaneously successful.

$P(T=n) = P(X_n = Y_n = 1) = (1-p_1p_2)^{n-1} p_1p_2$, $n = 1, 2, 3, \cdots$

$E[T] = \frac{1}{p_1p_2}$

(b) the two both failed : $(1-p_1)(1-p_2)$

let $q = (1-p_1)(1-p_2)$. $E = \frac{1}{1-q} = \frac{1}{p_1+p_2-p_1p_2}$

(c) Assume M : their first sucess are simultaneous.

$P(M=n) = [(1-p)^2]^{n-1} p^2$ $(p_1 = p_2 = p)$

$P(M) = \sum_{n=1}^{\infty} P(M=n) = p^2[1 + (1-p)^2 + (1-p)^4 + \cdots] = p^2 \cdot \frac{1}{1-(1-p)^2} = \frac{p^2}{p(2-p)} = \frac{p}{2-p}$.

Mario OR Zelda succeed first does not matter, so

$2p' + P(M) = 1$, $2p' = 1 - \frac{p}{2-p} = \frac{2-2p}{2-p}$.

$p' = \frac{1-p}{2-p}$

Problem 4

Show that for any events $A_1, \ldots, A_n$,

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) \geq \sum_{j=1}^{n} P(A_j) - n + 1.$$

Hint: First prove a similar statement about indicator r.v.s, by interpreting what the events $I(A_1 \cap A_2 \cdots \cap A_n) = 1$ and $I(A_1 \cap A_2 \cdots \cap A_n) = 0$ mean.

$I(A) = \begin{cases} 1, & A \text{ happens} \\ 0, & A \text{ doesn't happen} \end{cases}$

First to prove $I(A_1 \cap \cdots \cap A_n) \geq \sum_{j=1}^{n} I(A_j) - n + 1$.

① all event happen

left $= 1$, right $= n - n + 1$.

② at least one event does not happen

left $= 0$, $\sum_{j=1}^{n} I(A_j) \leq n - 1$., left $\geq$ right.

get the expected value from both sides.

$E[I(A_1 \cap \cdots \cap A_n)] \geq E[\sum_{j=1}^{n} I(A_j) - n + 1] = \sum_{j=1}^{n} E[I(A_j)] - n + 1$

As $E[I(A)] = P(A)$

$P(A_1 \cap \cdots \cap A_n) \geq \sum_{j=1}^{n} P(A_j) - n + 1$